



Carbon and water footprint assessment of the production cycle of the black soldier fly (*Hermetia illucens*) on a farm in Spain

J.J. Galán-Díaz^{a,*}, L. Pena-Mosquera^a, J. Puertas-Agudo^a, Jesús Rodríguez^b

^a ETSI Caminos, Canales y Puertos, University of A Coruña, A Coruña, Spain

^b CEO Bioflytech Group, Agricultural Engineer, Spain

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ABSTRACT

A comprehensive investigation into the carbon and water footprint of a black soldier fly (*Hermetia illucens*) farm in Murcia, Spain, was carried out following ISO 14064-1 standards. The study, utilizing data from BioflyTech, the managing company, spanned various departments to comprehensively analyze production and consumption. The results highlight that the carbon footprint associated with fresh larva production is approximately thirty times lower than that of conventional animal farming, aligning with sustainable practices and offering a promising alternative in the quest for eco-friendly protein sources. Moreover, the water footprint of the black soldier fly farm is significantly lower than that of traditional animal farms, underscoring the environmental advantages of this innovative approach to larva production over conventional livestock farming. These valuable insights contribute substantively to the ongoing discourse on sustainable and responsible agricultural practices, paving the way for more environmentally friendly solutions in the ever-evolving landscape of sustainable food production practices.

1. Introduction

The concept of carrying capacity, as defined by Cohen, refers to the number of individuals in a population that can be sustained by the resources of a given habitat, representing the asymptote or plateau in the logistic and other sigmoid equations used to model population growth (Cohen, 1995). Despite the challenges in calculating this capacity, some authors, such as Romero-Oliva, estimate that it may fall within a range of 6000 to 15,000 million inhabitants, with an average of about 10,000 million inhabitants (Romero-Oliva, 2023). This scenario is rapidly approaching, according to the UN's projections for the year 2050, where the global population is expected to reach 10 billion inhabitants (Nations, 2019).

This sharp population increase raises significant concerns about food supply for both humans and animals. The primary source of protein in diets is of animal origin, posing a sustainability challenge for the planet. Livestock production is estimated to occupy 30% of the world's land - excluding frozen surfaces - or 75% of agricultural land, and it consumes approximately 8% of global human water use (FAO (Food and Agriculture Organization of the United Nations), 2009).

Additionally, it should be noted that over 14% of the total CO₂ emitted into the atmosphere corresponds to this sector - 7.1 Gigatons of CO₂-equivalent per year (Gerber et al., 2013).

When examining global averages, it becomes apparent that the water footprint of meat increases incrementally from chicken meat (4300 m³/t), goat meat (5500 m³/t), pig meat (6000 m³/t), and sheep meat (10,400 m³/t) to beef (15,400 m³/t). These variations can

* Corresponding author.

E-mail address: juan.jose.galan@udc.es (J.J. Galán-Díaz).

be partly attributed to differences in the feed conversion efficiencies of these animals (Mekonnen and Hoekstra, 2012).

Regarding water footprint analysis, Mekonnen et al. (Mekonnen and Hoekstra, 2012) highlight that beef cattle have the largest contribution, accounting for 33% of the total water footprint, followed by dairy cattle and pigs, each representing 19% of the total water footprint. According to these authors, the total water footprint associated with animal production amounts to 2422 Gm³/year.

In contrast, chicken products (including chicken meat and eggs) in the industrial production system generally exhibit a lower blue and grey water footprint per ton of product compared to the other two production systems. This is primarily due to the predominant use of concentrated feed in chicken production, whether intensive or extensive. This difference underscores the relative water use efficiency of poultry production compared to other forms of animal production. However, a detailed analysis of chicken meals may reveal more nuances about their specific impact on the global water footprint.

Due to all the aforementioned factors, it becomes of vital importance to find a protein source that has high nutritional value while also being environmentally friendly in terms of greenhouse gas emissions and water consumption. Insect rearing for protein production is a viable alternative both economically and environmentally. The speed at which insects reach maturity and the number of offspring generated by each individual is much higher than that of any farm animal.

The use of insects in the human diet is common in some parts of the planet, such as Asia, Latin America, or Africa (Makkar et al., 2014). Recently, the European Commission has approved the consumption of the house cricket (*Acheta domestica*) (European Parliament and the Council of the European Union, 2011). However, Europeans are still hesitant to include insects in their regular diet.

Indeed, insect larvae, such as the black soldier fly (*H. illucens*), have been increasingly used to produce flour and feed for animal nutrition due to their nutritional qualities. Barragan-Fonseca et al. (Barragan-Fonseca, Dicke and van Loon, 2018) investigated the effects of nutrient concentration in the diet and larval density on the performance and body composition of *Hermetia illucens* larvae. In their study, two feeding regimes were employed: one visually adjusted and another with fixed ad libitum ration. They found high larval survival rates under both regimes (approximately 95%), yet larval density and nutrient concentration significantly influenced development time and individual larval weight. For instance, larvae reached an average weight of 0.18 ± 0.03 g when reared at a density of 250 larvae per container with a high-protein, high-fat diet. Furthermore, body composition analysis revealed these larvae had higher protein content ($55.7 \pm 0.9\%$ dry weight) and fat content ($32.4 \pm 1.0\%$ dry weight). These findings underscore the importance of optimizing diet nutrient quality and managing larval density to enhance both the performance and nutritional quality of *H. illucens* larvae, critical for their potential application in large-scale food and feed production systems. In addition to its nutritional value, potential medical applications of *Hermetia illucens* have also been investigated. Guarnier et al. (Guarnieri et al., 2022) explored the antimicrobial properties of chitosan derived from larvae, pupal exuviae, and deceased adults of this species. Using Fourier-transform infrared spectroscopy, they confirmed the structural similarity of the produced chitosan to commercial counterparts and found it exhibits significant antimicrobial activity. These findings open new avenues for utilizing this biopolymer in various biological and medical fields.

Several recent studies have assessed the feasibility of substituting black soldier fly larvae meal (BSFLM) for fish meal (FM) in broiler diets. Stejskal et al. (2023) found that incorporating BSFLM up to 18% did not affect growth parameters in European perch. Marin et al. (2023), through a meta-analysis of 18 studies involving 4229 broiler chickens, concluded that diets containing up to 15% BSFLM showed no detrimental effects on growth rates. Nampijja et al. (2023) investigated higher inclusion levels and determined that substituting up to 500 g/kg of BSFLM reduced feed intake, improved feed conversion ratio (FCR), and maintained weight gain in broilers. They found optimal substitution at 540 g/kg based on FCR, despite slight decreases in muscle protein and Omega-3 fatty acids. These studies collectively suggest that replacing FM with BSFLM up to 540 g/kg is viable, providing economic benefits without compromising broiler growth or meat quality (Nampijja et al., 2023).

In this paper, an environmental study of a black soldier fly farm for the production of meal, fat, and compost located in the Region of Murcia (Spain) is conducted. It should be noted that several studies have been conducted on the environmental impact of *Hermetia illucens*: Mertenat et al. (2023) (Mertenat et al., 2019) explored the potential for mitigating greenhouse gas emissions through black soldier fly (BSF) larval waste treatment in Indonesia, comparing it with open windrow composting. Using a life cycle assessment (LCA) approach, they demonstrated that direct CO₂eq emissions at the BSF facility were significantly lower compared to composting, being up to 47 times lower. The study identified that the main contributors to global warming potential (GWP) were post-composting residues (69%) and electricity consumption (up to 55%), highlighting the potential substitution of fish meal with BSF larvae meal as a strategy to reduce GWP by up to 30%.

Wancheng Pang et al. (Pang et al., 2020) investigated the bioconversion of food waste by BSF larvae, assessing how different pH conditions influenced carbon and nitrogen conversion efficiency, as well as emissions of gases such as CO₂, NH₃, CH₄ and N₂O. Their findings underscored that higher pH in substrates enhances larval production and leads to significant reductions in greenhouse gas emissions, particularly CH₄ and N₂O. This study highlights the potential of BSF treatment not only for sustainable waste management but also as an effective climate mitigation strategy in urban contexts of low and middle-income countries.

Besides, Kofi Armah Boakye-Yiadom et al. (Boakye-Yiadom et al., 2022) conducted a comprehensive review focusing on evaluating the greenhouse gas emissions and conducting a life cycle assessment (LCA) of black soldier fly (BSF) larvae biotreatment activities. Their study underscores the environmental performance of BSF-derived products in mitigating challenges related to feed and organic waste management. The review discusses various gas sensing techniques used to quantify GHG and ammonia emissions during BSF bioconversion processes, emphasizing the need for standardized guidelines to enhance methodological approaches in impact assessments.

The study has been carried out following the most recent methodologies for calculating carbon footprint and water footprint in order to obtain reliable and realistic results.

The investigation into the environmental impact of black soldier fly farming is critical, considering the current global challenges in

population growth and the associated strain on traditional protein sources. As the world hurtles toward a projected population of 10 billion by 2050, the need for sustainable and efficient protein production methods becomes paramount. Livestock production, a major contributor to environmental degradation, demands vast land and water resources while emitting significant amounts of greenhouse gases. The quest for alternative protein sources has led to a growing interest in insects, particularly the black soldier fly, as a promising solution. Despite insect consumption being commonplace in certain regions, the acceptance and integration of insects into Western diets remains a hurdle.

The black soldier fly larvae's nutritional composition, including high protein content and potential natural antibiotics, positions them as a compelling option for animal feed and, potentially, human consumption. Various studies have explored the feasibility of incorporating black soldier fly larvae meal into animal diets, revealing positive outcomes without compromising growth parameters. This paper addresses a notable gap in scientific literature by conducting an environmental analysis of a black soldier fly farm in the Region of Murcia (Spain).

The study adheres to state-of-the-art methodologies for calculating carbon and water footprints, aiming to provide reliable and realistic insights into the environmental impact of black soldier fly farming. The outcomes of this investigation contribute valuable information to the discourse on sustainable protein sources amid the escalating challenges posed by global population growth.

2. Materials and methods

2.1. Description of the farm and production cycle

The company that manages the black soldier fly farm is BioflyTech (hereinafter referred to as BT), located in the Region of Murcia (Spain), in a facility covering an area of 21,000 m². The company BT provides its customers with protein flours, functional fats, and enriched compost. All these products are made from the breeding of dipteran insects of *Hermetia illucens*.

The productive cycle of the black soldier fly on the farm consists of four stages:

Rearing stage (Hatchery): Part of the larvae from the eggs of *Hermetia illucens* will be used to generate new adult flies, which, in turn, will lay the eggs that will allow the maintenance of the black soldier fly production process.

Fattening-Dehydration Stage: The majority of the neonates are allocated to larvae production. These larvae (those that were not used in the previous stage) undergo a fattening process, which is the most critical part of the production. Additionally, the option of drying and freezing some of the larvae is made possible in this stage.

Rendering Stage: Upon reaching the expected final weight (0.1–0.11 g per unit) of the larvae, the fattening phase is completed. During this rendering stage, the final products, such as flour and fat, are obtained from the dried larvae.

Post-Production and Sales Stage: In this stage, the products are packaged for sale and stored until they are effectively shipped to customers.

In each of these stages, the company BT obtains various products aimed at their commercialization:

1. Products from the Hatchery Stage:

- a. *Hermetia illucens* eggs
- b. *Hermetia illucens* larvae (neonates)
- c. Puparium fertilizer. The fertilizer is a product derived from the mixture of the substrate (where larval fattening takes place, essentially feed) with the larval waste until they transition into the adult stage. This fertilizer, which includes the remains of the puparium, has a higher added value

2. Products in the Fattening Stage:

- a. Fresh *Hermetia illucens* larvae
- b. Frozen *Hermetia illucens* larvae
- c. Dried *Hermetia illucens* larvae
- d. Fertilizer. This product does not contain remains of the puparium; it is a byproduct derived from the larval fattening substrate

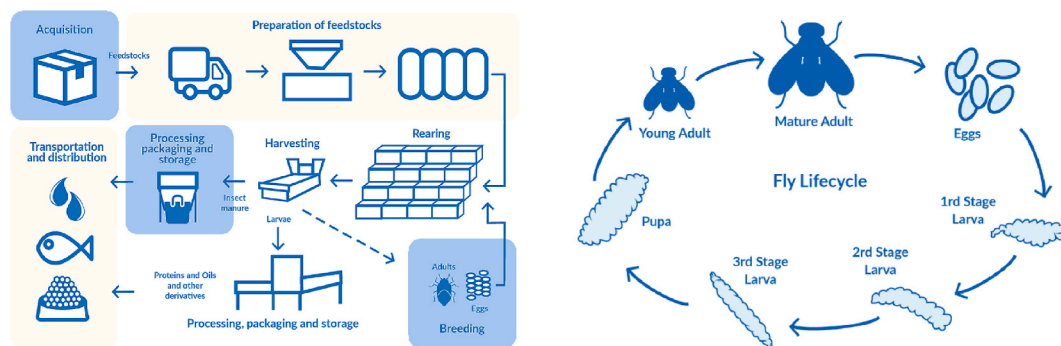


Fig. 1. Schematic diagram of BT farm production system and black soldier fly life cycle.

3. Products from the *Final Rendering Stage*:

- a. Fats - Two types of fats are included based on the extraction method: “gg” fats are obtained through a traditional cold-pressed system, while “eg” fats are extracted through a gas extraction method
- b. Flours - The flours are also subdivided following the same criteria indicated for the fats

Fig. 1 displays a basic outline of the life cycle and production of flies, as per the International Platform of Insects for Food and Feed (IPIFF) ([IPIFF-Guide-on-Good-Hygiene-Practices.pdf](#)).

2.2. Carbon footprint

The methodology for calculating Greenhouse Gas (GHG) emissions at BT is based on ISO 14064-1 standard and considers the following aspects: Principles of GHG accounting and reporting; determination of organizational boundaries; determination of operational boundaries; tracking emissions over time; and identification and calculation of GHG emissions.

According to ISO 14064-1, the following scopes can be distinguished:

Scope 1: Direct GHG emissions. Scope 1 emissions are considered emissions from sources that are owned or controlled by the company.

Scope 2: Indirect GHG emissions associated with electricity and heat. Scope 2 emissions are indirect emissions resulting from the company's activities but occur from sources not owned or controlled by the company.

Scope 3: Other indirect emissions.

The company BT compiles its emissions through the control approach, accounting for 100% of the GHG emissions attributable to the operations of the process over which it exercises control. Regarding operational boundaries, it is noted that the company accounts for at least scopes 1 and 2 separately.

In BT, indirect GHG emissions originating from electricity generation have been quantified. The electricity consumption is proportionally allocated across all stages of the studied process. Additionally, the substrate-feed used for the breeding and fattening of flies has also been considered as a scope 2 emission. BT does not generate indirect emissions that need to be considered in this study.

2.2.1. Exclusions and baseline year

The direct and indirect GHG sources whose contribution to emissions is not significant (less than 5% of the total emissions) and those whose quantification is not technically feasible or cost-effective have been excluded from this study. In this regard, direct GHG emissions from the drying kilns were excluded.

The baseline year inventory can be used to monitor the performance of the process by BT towards certain emission objectives, in which case the baseline year is referred to as the target baseline year. The year 2021 has been chosen as the target baseline year.

2.2.2. Quantification methodology for emissions

For the present study, the GHG emissions of the company BT have been quantified and documented according to the following phases: *Identification of emission sources*, *Selection of quantification methodology*, *Selection and collection of GHG data from the process*, *Selection or development of GHG emission factors* and *Calculation of GHG emissions*.

2.2.3. Emission sources

The GHG sources of BT contributing to direct GHG emissions (Scope 1), and indirect emissions (Scope 2) have been identified and documented separately. The GHG emissions come from the following source categories: *Substrate consumption for fly feeding*, *Electricity consumption*, *Water consumption*, *Diesel consumption* and *Labor force utilization*.

Table 1 displays the production data for the reported year provided by the company through its Quality, Production Engineering, and Finance departments.

2.3. Emission factors

The emission factors express the kilograms of CO₂ pollutant emitted into the atmosphere per corresponding unit of electricity, feed, diesel, water, and labor.

Table 1
Annual production values of different farm products.

Annual Production	Total (kg)
Eggs	2880
Neonate	1440
Fresh Larvae	4,000,000
Dried Larvae	1,400,000
Fat Material	445,600
Flour	953,400
Pupal Fertilizer	60,754
Fertilizer	3,015,812

Therefore, to calculate greenhouse gas emissions, the following expression will be used:

$$CO_2 \text{ emissions} = \text{Consumption data} \times \text{Emission factor} \quad (1)$$

The emissions related to electricity (Scope 2) have been calculated considering the electricity mix factor provided by the Spanish Electrical Grid for the year 2021, with a value of 0.14 tCO₂eq/MWh (Inicio | Red Eléctrica (ree.es)). For the substrate-feed used in the fattening process, the emission factor has been obtained by considering the mass percentage of each cereal that makes it up, using the values reported by researchers in specialized publications, and its value was 0.38 kgCO₂eq/kg (Mekonnen and Hoekstra, 2012). The emission factor for diesel is the one indicated by the Ministry for Ecological Transition (<https://www.miteco.gob.es/es/>): 2.79 kg CO₂ eq/L. The emission factor for water is the one provided by Aguas de Murcia (Emuasa) (<https://www.emuasa.es/>): 0.059 kg CO₂ eq/m³. And finally, to determine the emission factor for labor force, it was estimated that the operator travels an average total distance of 10 km daily in a diesel motor vehicle consuming 5 L per 100 km and works approximately 8 h per day. The emission factor for diesel was used as the starting value (Mekonnen and Hoekstra, 2012): 0.172 kg CO₂ eq/hora.

2.4. Water footprint

To determine the water footprint of BT's products, the allocation of all farm water consumption is essential. Each final product is assigned a share of water based on its selling price. This method ensures a nuanced distribution, attributing a greater responsibility in the water footprint to products with higher final values in the sales balance. This meticulous approach aligns with BT's commitment to transparent and comprehensive environmental assessments.

In the analysis of the water footprint, the sum of green water consumption (rainwater), blue water consumption (water from the network), and grey water consumption (water polluted) will be considered.

The conversion factors for calculating the water footprint have been obtained as follows: the one for electricity was provided by the electricity supplier itself (Inicio | Red Eléctrica (ree.es)), and the one for animal feed was determined by considering the conversion factors for each cereal it contains, as per specialized literature (Mekonnen and Hoekstra, 2012). This took into account the mass percentage of each cereal in the feed. In the case of the water footprint of water consumed on the farm, the supplier's guidelines recommend considering an additional 20% over the actual consumption amount (<https://www.emuasa.es/>). As for labor, the same assumption was made as in calculating the carbon footprint, but with the consideration of emitting 5.5 kg of CO₂ for every 40 km traveled. To calculate the water footprint of diesel fuel, we used data from Vanham et al., which provides a value of 249 m³ TJ⁻¹ for energy generated from petroleum (Vanham et al., 2019). Additionally, we considered information from Hofstrand, who stated that 1 L of diesel is equivalent to 3.5×10^{-5} TJ (Hofstrand, 2007). Both the conversion factors for the carbon footprint and the water footprint are presented in Table 2.

3. Results and discussion

3.1. Carbon footprint

Considering equation (1) and the consumption data provided for the reported year by the company through its quality, production engineering, and finance departments, the total emissions have been calculated for each emitting source, as well as the total CO₂ equivalent emissions from the farm. The results are presented in Table 3.

The previous data can be visualized as percentages in the sector diagram shown in Fig. 2.

In accordance with Fig. 2, the substrate-feed emerges as the most substantial source of pollution in the production of the farm process, primarily due to the significant quantities employed in its formulation.

Beyond computing the total annual emissions of the farm, a careful breakdown of the annual emissions linked to each product manufactured by the BT company for sale has been undertaken. This entails considering both the quantity sold for each product and its publicly listed selling price. This comprehensive evaluation aims to derive a nuanced understanding of the partial weighting for each product concerning the overall sales volume, facilitating the estimation of the emissions associated with each specific product based on the total farm emissions.

To achieve this, the quantity sold for each product is factored in, along with its public selling price, yielding a comprehensive insight into the emission contribution of each product. This quantity is then divided by the weight of each product sold in kilograms, resulting in the final emission parameter per kilogram of product (kg CO₂ eq emission/kg of product), as elucidated in the last column of Table 5.

Table 2
Emission factors for each emission source.

Emission Factor	kg CO ₂ eq/unit.	m ³ H ₂ O/unit
Electricity Consumption	0.12 kwh	0.00481
Feed Consumption	0.380 kg	1.444
Water Consumption	0,059 m ³	1.200
Labor force	0.172 h	0.00625
Diesel Consumption	2.721 L	0.00890

Table 3
Emission sources in kg CO₂ equivalent and total emissions of the farm.

Concept	Unit	Consumptions	kg CO ₂ eq
Feed	kg	5,026,353	1,910,014
Electricity	Kwh	702,720	84,326
Water	m ³	8467	500
Labor force	hours	78,320	13,471
Diesel	L	145,674	396,380
			2,404,691

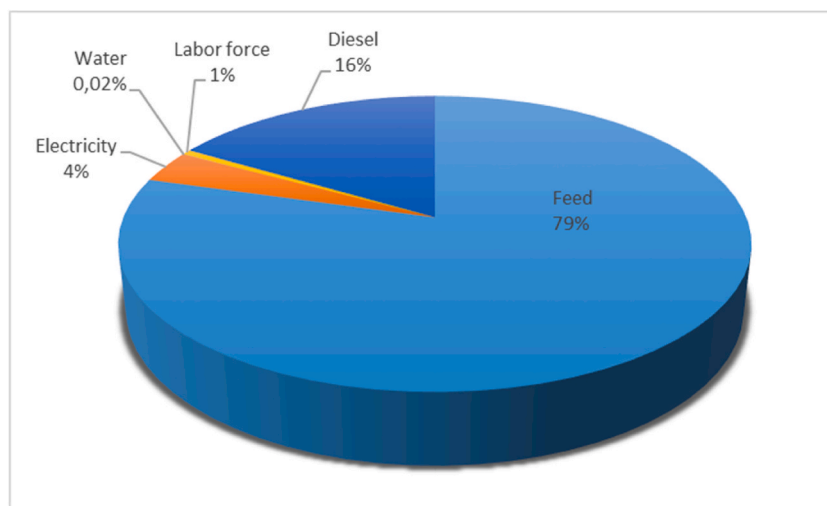


Fig. 2. Distribution of CO₂ emissions to the atmosphere according to the emission source.

It's noteworthy that the 15% attributed to egg sales signifies that 85% of the production is internally utilized within the hatchery to complete the reproductive cycle. Conversely, a 0% value in the sales column indicates that, in the base year 2021, no sales of that particular product were made, whether due to production constraints, operational considerations, or commercial factors (in this case, fresh larvae, frozen larvae, and dried larvae).

This detailed information is presented in Table 4 in the Appendix. It provides a comprehensive breakdown of the carbon footprint associated with each product sold.

In the prevailing circumstances, where no fresh *Hermetia illucens* larvae have been marketed, it becomes imperative to assume that the entire production of fresh larvae was indeed sold in its entirety. This assumption facilitates the estimation of the carbon footprint solely for this product, without considering any derivatives such as dried larvae, fat material, or flour. Consequently, the complete carbon footprint of these products is attributed solely to the fresh larvae.

Under this assumption and considering an average selling price of 1.6 Euros per kilogram of fresh larvae throughout the year, the aggregate revenue for the entire production would amount to 6,400,000 euros. This revenue contribution corresponds to approximately 84% of the total farm sales, thereby resulting in a maximum emission value of 0.506 kg of CO₂ eq per kilogram of fresh larvae.

In light of the emission values for various protein sources, where the carbon footprint of White Pork is 3.42, chicken is 3.76, lamb is 2.1, beef is 15.4, fish is 4, and eggs are 2.90 kg CO₂ eq/kg, a discerning inference can be drawn. The production of protein derived from *Hermetia illucens* fly larvae is approximately thirty times lower than that originating from beef and about five times lower than that derived from avian eggs (Treu et al., 2017). This nuanced comparison underscores the significant environmental advantages associated with the production of protein from *Hermetia illucens* fly larvae in comparison to traditional protein sources.

3.1.1. Calculation of emissions by activity

To discern the most pivotal and influential productive activities contributing to greenhouse gas emissions, an in-depth analysis of the carbon footprint associated with various activities within the production process has been thoroughly undertaken. This evaluative exercise serves as a foundational step for the company, guiding the formulation and implementation of strategies aimed at bolstering efficiency in decarbonization and consequent reduction of greenhouse gas emissions.

Upon comprehending the intricacies of the production process visually depicted in Fig. 1 and duly considering the percentage of consumption from the primary emission sources in each activity, a precise calculation of the carbon footprint ensued. The outcomes of this assessment are methodically presented in Table 5 (Appendix).

Upon grouping the values associated with feed, a conspicuous observation emerges – more than 75% of the carbon footprint is attributed to feeding activities. This salient insight is illustrated in the graphical representation presented in Fig. 3. This discernment

serves as a foundational guidepost for strategic interventions, emphasizing the need for targeted measures to optimize efficiency and curtail emissions in the feeding aspects of the production process.

3.2. Water footprint

In the BT company, water consumption originates from two sources: directly within the factory during various production activities, including water supplied to feeding trays for larval development, tracked through water billing data as blue water usage. The second source involves water indirectly provided to larvae through their diet, predominantly sourced from the water used in cereal production.

As indicated in Table 1, the water directly consumed from the municipal water supply network amounts to 8,467,000 L per year.

The water indirectly provided can be estimated from the cereals that make up the feed consumed by the flies. These cereals have a high content of green water, as they are typically rain-fed crops. In Table 4, the cereals that make up more than 95% of the diet of BT farm flies are listed, along with the volume of water consumed in their production. Blue water is from surface and groundwater sources used for consumption and irrigation; green water is rainwater absorbed by plants; and grey water is domestic wastewater without fecal contamination, reusable after treatment. In Table 4, it is evident that barley, wheat, corn, and beetroot, which constitute the diet of *Hermetia illucens* flies, utilize different volumes of green, blue, and grey water, with barley and wheat being the primary components.

Drawing insights from the data presented in Table 5, an average water footprint of 1444 cubic meters per ton (m^3/t) has been employed as the representative measure for the water footprint of the feed.

The intricate process of allocating the water footprint to each product involves a comprehensive consideration of the overall water consumption, encompassing both direct usage and the water embedded in the feed. This allocation is methodically distributed among the diverse products manufactured by BT. This intricate procedure unfolds in two distinct phases: initially, products such as eggs, larvae, and fertilizer are individually contemplated; subsequently, an in-depth analysis of the larvae's breakdown into more refined products is undertaken. The detailed breakdown of water consumption for each farm input, coupled with their respective percentage contributions, is presented in Table 5. This approach ensures a thorough examination of water utilization across various stages of production.

Utilizing the aggregate water consumption data encompassing all activities on the farm, it becomes imperative to allocate a distinct water footprint to each product by considering the sales volumes of each product. The rigorous calculation of the water footprint for specific products, such as eggs, neonates, fresh larvae, and fertilizer, is presented comprehensively in Table 8 (Appendix). This approach ensures a nuanced understanding of the water-related impact associated with each facet of the production of the farm, allowing for a granular assessment of resource utilization and environmental responsibility.

Under the assumption of identical conditions as those stipulated for the carbon footprint analysis, where all fresh larvae are commercially sold without deriving any by-products, a water footprint of 1.529 cubic meters of water per kilogram of fresh larvae is carefully computed. A comparative examination against the water footprints of conventionally consumed meats like chicken, pork, sheep, and beef, registering at 4.325, 5.988, 8.763, and 15.415 cubic meters per ton, respectively (Vauterin et al., 2021), unequivocally reveals the markedly reduced water footprint associated with *Hermetia illucens* larvae produced by BT. This disparity is particularly striking in the case of beef, where the water footprint surpasses tenfold.

It is imperative to note that fresh larvae are not the typical presentation form for BT products intended for consumption. Following a drying process that expels 65% of the larva's mass in the form of water (a percentage analogous to that of traditional farm animals), the end result is dried larvae. This form enhances ease of preservation, storage, and transportation.

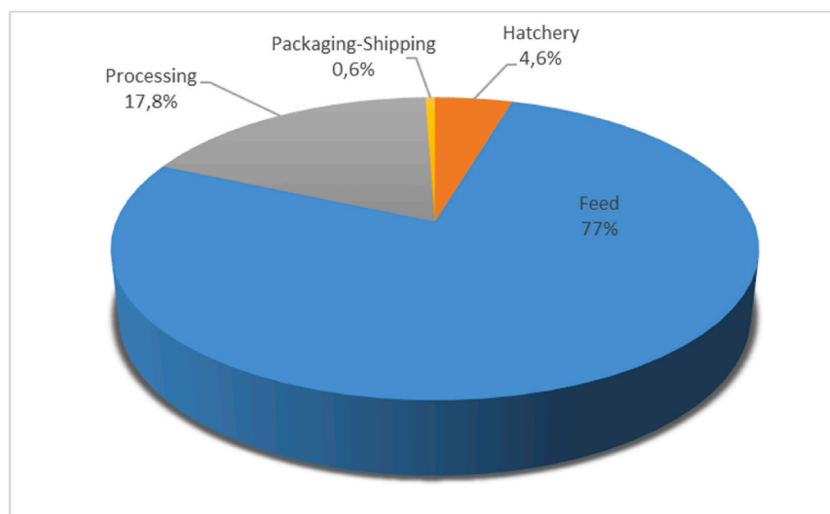


Fig. 3. Distribution of CO₂ emissions into the atmosphere based on the emission source for each activity of BT company.

Table 4

Percentage of each cereal in the diet of *Hermetia illucens* flies and the volume of each type of water consumed in their production.

Cereal	Percentage in the diet	Green	Blue	Grey	Total
Barley	48.07	1213	79	131	1423
Wheat	24.37	1277	342	207	1826
Corn	19.83	947	81	194	1222
Beetroot	3	82	26	25	133

Table 5

Water consumption by farm inputs and percentage contribution.

Water footprint source	Total H ₂ O (m ³)	Percentage (%)
Feed	7,258,054	99.8
Electricity	3380	0.046
Water	10,160	0.140
Labor force	490	0.0067
Diesel	1297	0.0178
	7,273,380	100

A noteworthy observation arises when considering highly refined protein flours, revealing a diminished water footprint (equivalent, in the case of chicken) in comparison to raw meat sourced from traditional livestock farming practices (Mekonnen and Hoekstra, 2012). This observation underscores the efficiency and sustainability embedded in the production processes of BT, particularly when compared to conventional methods prevalent in the agricultural industry.

4. Conclusions

A comprehensive investigation has been undertaken to analyze the carbon and water footprint of a black soldier fly (*Hermetia illucens*) farm in detailed managed by BioflyTech. Situated in the expansive expanse of 21,000 square meters in the Murcia region of Spain, this study adheres to the stringent guidelines outlined by the ISO 14064-1 standard. The examination encompasses a holistic approach, delving into both the overall emissions generated by the farm and the emissions intricately associated with each distinct activity and product earmarked for commercialization.

Data pertinent to consumption and production on the farm were diligently collated from diverse sources, with contributions from the quality, production, engineering, and finance departments of BioflyTech. These departments, each playing a pivotal role in the operations of the farm, have provided essential insights into the intricacies of resource utilization and product generation. The collaboration among these sectors ensures a comprehensive and multifaceted understanding of the farm's environmental impact.

Furthermore, the emission factors utilized in this study, a critical component for accurate assessment, have been diligently sourced from corresponding entities with expertise in emission quantification. These factors, rooted in scientific rigor, enhance the precision of the evaluation by incorporating industry-recognized metrics and benchmarks. The collaboration between BioflyTech and these external entities underscores the commitment to methodological rigor and accuracy in the determination of both carbon and water footprints.

By embarking on this thorough analysis, BioflyTech endeavors not only to quantify its environmental impact but also to establish a benchmark for responsible and sustainable farming practices within the burgeoning sector of black soldier fly cultivation. This study aims to contribute valuable insights to the broader discourse on eco-friendly agricultural methodologies, emphasizing the imperative of balancing production demands with ecological stewardship.

The main conclusions of the study are as follows:

The cumulative emissions from the farm amount to 2,404,691 kg of CO₂eq, of which nearly 80% emanate from the production process of fly feed.

The specific operational activity denoted as 'Feeding Neonate to Larva - Larval Seed - Larva - Adult' emerges as the predominant contributor to the overall carbon footprint within the spectrum of the farm's activities, constituting more than 50% of the aggregate CO₂ emissions.

In the sphere of protein production derived from *Hermetia illucens* larvae, the emissions are approximately thirty times lower than those stemming from cattle (beef) and five times less than those associated with chicken egg production.

The comprehensive water requirement for the entire spectrum of farm products stands at 7,273,380 cubic meters.

The water footprint attributed to the production of fresh larval protein is quantified at 1.293 cubic meters per ton, representing a markedly lower impact compared to conventional meat proteins commonly consumed. This emphasizes the sustainable advantage of larval protein production in mitigating water-related environmental concerns.

This in-depth analysis not only quantifies the environmental impact of the farm but also underscores the significant role that alternative protein sources, such as *Hermetia illucens* larvae, can play in fostering sustainable agricultural practices. The utilization of such data is pivotal for steering agricultural enterprises towards ecologically conscious decisions and contributing to the ongoing discourse on sustainable food production.

CRedit authorship contribution statement

J.J. Galán-Díaz: Writing – review & editing, Data curation, Conceptualization. **L. Pena-Mosquera:** Writing – original draft, Methodology, Data curation. **J. Puertas-Agudo:** Supervision, Software, Data curation. **Jesús Rodríguez:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data is included within the paper itself or its appendix.

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Appendix

Tables 4, 5, and 8.

Table 4

Weighting factor of all farm products intended for sale, CO₂ emission quantity for each product, and carbon footprint of each product sold.

Product	Production (kg)	Sales (%)	Retail Price* (€)	Total-Sales (€)	Weighting Factor	Emisión CO ₂ (kg)	kg CO ₂ eq/kg
Eggs	2880	15	1250	540,000	0.12973	312,993	722.151
Neonate	1440	15	1250	270,000	0.06487	155,985	722.151
Fresh Larvae	4,000,000	0	1.60	0	0.0000	0.000	0.000
Dried Larvae	1,400,000	0	2.50	0	0.0000	0	0.000
Fat Material	445,600	100	1.02	454,512	0.10920	262,581	0.589
Flour	953,400	100	2.62	2,497,908	0.60012	1,443,094	1.514
Fertilizer	3,076,566	1	0.13	399,954	0.09609	231,062	0.075

(*) The value indicated in this column corresponds to an annual average of the selling price.

Table 5

Emission by activity.

Activity	Animal feed	Electricity.	water	Labor	Diesel	kg CO ₂ eq/kg
Hatchery from Adult to Egg	5	15	5	15	0	110,195
Feeding Egg to Neonate	20	10	15	10	0	391,858
Feeding Neonate to Larvae - Seed Larvae - Larvae - Adult	70	15	15	20	0	1,352,428
Feeding Larvae to Fresh Larvae	5	10	55	30	0	108,249
Freezing Process of Larvae	0	0	0	0	0	0
Drying Process	0	20	4	15	100	415,286
Rendering of Fresh Larvae	0	0	0	0	0	0
Rendering of Fresh-Frozen Larvae	0	0	0	0	0	0
Rendering of Dried Larvae	0	15	2	5	0	13,333
Packaging and Storage	0	10	2	3	0	8847
Shipping and Sale	0	5	2	2	0	4495
TOTAL	100	100	100	100	100	2,404,691

Table 8

Water footprint for the final products of the BT farm.

Product	Production (kg)	Sales (%)	Retail Price (€)	Total Sales (€)	Weighting Factor	water (m ³)	Water footprint (m ³ H ₂ O/kg)
Eggs	2,880	15	1,250	540,000	0.12973	943,602	2,184
Neonate	1440	15	1,250	270,000	0.06487	471,801	2,184
Fresh larvae	4000000	0	1.60	0	0.00000	0	0
Dried-larvae	1,400,000	0	2.50	0	0.00000	0	0
Fat material	445,600	100	1.02	454,512	0.10920	794,220	1.782
Flour	953,400	100	2.62	2,497,908	0.60012	4,364,874	4.578
Fertilizer	3,076,566	100	0.13	399,954	0.09609	698,884	0.277

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